

Charge perturbations in Q: the formulas

SCAAS polarization restraints · the Born artifact · a thermodynamically consistent correction

Part A – SCAAS polarization restraints (Warshel & King, 1985)

A1 • Three-region partition of the potential – Eq. (1)

$$V_{\text{tot}} = V^{\text{aa}} + V^{\text{as}} + V^{\text{ss}} + V^{\text{sb}} + V^{\text{ab}} + V^{\text{bb}}$$

Regions: (a) solute + inner shells, (s) surface shell, (b) bulk (implicit). Only (a+s) is simulated explicitly; $V^{\text{ab}}, V^{\text{bb}}, V^{\text{sb}}$ become effective terms. V^{sb} carries the *surface constraints*; V^{ab} is the continuum (Born-type) term.

A2 • The surface-constraint force – Eq. (4)

$$f_{j,n}^{\text{sb}} = -\left(K_0 + \Delta K_n\right)\left[Y_j - (\langle Y_j^0 \rangle + \Delta Y_j^n)\right]h_j - \mu(\gamma_0 + \Delta\gamma_n)\dot{Y}_j + A_n(t)$$

Y_j is a surface variable (radial distance R or polarization angle θ), sorted by magnitude each step. A harmonic restraint toward the bulk reference $\langle Y_j^0 \rangle$, plus Langevin friction and random force. A soft restraint, not a hard constraint.

A3 • Radial reference from the bulk RDF – Eq. (11)

$$\int_0^{\langle R_j^0 \rangle} g(R) 4\pi\rho R^2 dR = (n_0 - 1) + j$$

Sets target shell radii so the integrated *bulk* radial distribution $g(R)$ places the right number of molecules inside each shell $\Rightarrow$ uniform, bulk-like density all the way to the boundary (the "density" restraint).

A4 • Angular reference $\Rightarrow$ the ideal $\sin\theta$ law – Eqs. (12)–(13)

$$\frac{n_s}{2}\left(1 - \cos\langle\theta_j^0\rangle\right) = j \quad \Longrightarrow \quad \langle\theta_j^0\rangle = \arccos\left(1 - \frac{2j}{n_s}\right)$$

For a *field-free* (neutral) system the dipole-vs-radial angle obeys $P(\theta) \propto \sin\theta$ (isotropic orientations). This is the bulk-like surface; without it, surface waters over-orient tangentially (the "polarization" restraint).

A5 • Charged-solute polarization target – King Eq. (22)

$$\theta_i^0 = \theta_i^{0,*} - \frac{3}{2} L(X) \sin\theta_i^{0,*}$$

$\theta_i^{0,*}$ is the field-free target and $L(X)$ is the Langevin response to the enclosed field. Q implements the same form as $\theta_i^0 = \theta_i^{0,*} - \frac{3}{2}c\sin\theta_i^{0,*}$, with $c \propto Q/r_{\text{shell}}^2$. Thus a charge perturbation changes the restraint Hamiltonian itself.

Part B – The Born artifact & Q’s boundary (Marelius et al., 1999)

B1 • The multistate mapping potential – Eqs. (2)–(3)

$$V_m = \sum_{i=1}^N \lambda_i^m V_i, \quad \sum_{i=1}^N \lambda_i^m = 1$$

$$\Delta G_{\text{initial} \rightarrow \text{final}} = \sum_{m=1}^{n-1} \Delta G_{m \rightarrow m+1}$$

Q does not jump $V_1 \rightarrow V_2$; it samples a chain of *effective* Hamiltonians, each a weighted blend of *all* N state potentials (weights sum to 1), and sums ΔG over the small windows. The superscript m is a window *label*, not a power. A 2-state FEP is just the mapping vector $(1 - \lambda, \lambda)$, i.e. $V(\lambda) = (1 - \lambda)V_1 + \lambda V_2$. States are evaluated *separately*; the mapping vector only sets how they combine for the forces – which is why a per-state term (C3) is well defined.

B2 • The FEP master equation – Eq. (1)

$$\Delta G_{i \rightarrow j} = -RT \ln \left\langle \exp[-(V_j - V_i)/RT] \right\rangle_i$$

The quantity we compute, evaluated between adjacent windows. Everything downstream is about making V_i and V_j – including their boundary contributions – comparable, so that only the physical change survives.

B3 • Q’s radial water restraint – Eq. (8)

$$V_{\text{water}}(r) = \begin{cases} \frac{1}{2} k_{\text{rad}} (r - r_0)^2 - D_e, & r > r_0 \\ D_e (1 - e^{a(r-r_0)})^2 - D_e, & r \leq r_0 \end{cases}$$

Q’s implementation of the SCAAS *density* restraint: a Morse attraction pulls waters toward the surface, a half-harmonic wall keeps them in $\Rightarrow$ uniform density to the boundary.

B4 • The ideal boundary polarization Q restrains to

$$p(\theta) = \frac{1}{2} \sin \theta$$

Q’s implementation of the SCAAS *angular* restraint (three subshells, 0.5/1.0/1.5 Å): each boundary water is rotated toward its target angle in this ideal distribution. Same physics as A4; fixes surface over-polarization.

B5 • The Born term – the finite-sphere artifact

$$\Delta G_{\text{Born}} = -\frac{1}{4\pi\epsilon_0} \frac{Q^2}{2r} \left(1 - \frac{1}{\epsilon}\right)$$

The continuum free energy of polarizing the medium *outside* the sphere by the enclosed net charge Q – the leading part of V^{ab} the explicit droplet does not contain. Scales as Q^2 : it cancels only when the two systems have equal radius r *and* equal net charge Q .

B6 • Continuum polarization around a charge – Fig. 1c

$$P(r) = \frac{Q}{4\pi r^2} \left(1 - \frac{1}{\varepsilon}\right)$$

The radial polarization profile the surface restraints aim to reproduce. It connects the *angular* restraint (A4/B4) to the same monopole response that gives the Born term (B5), but the restrained explicit shell does *not* by itself supply the missing continuum energy outside the sphere.

B7 • Coulomb correction for neutralized ionic sites – Eq. (9)

$$\Delta G_{\text{corr.}}^{\text{el}} = \frac{1}{4\pi\varepsilon_0} \sum_{\substack{p \in \text{neglected sites} \\ l \in \text{ligand}}} \frac{q_p q_l}{\varepsilon r_{p-l}}$$

Charged residues near the boundary are modelled as neutral dipoles (they would be screened by high ε in reality). This adds back the ligand–(neutralized site) Coulomb energy with a high $\varepsilon \approx 80$. The size/charge-matching bookkeeping that makes the two legs comparable.

Part C – The bridge: charge-perturbation corrections (this work)

C1 • The finite-sphere Born self-energy, made concrete

$$E_{\text{Born}}(Q) = -C Q^2, \quad C = \frac{k_e (1 - 1/\varepsilon)}{2R}$$

The same Born term as B5, in Q's units: $k_e = 332.0637 \frac{\text{kcal}\cdot\text{\AA}}{\text{mol}\cdot\text{e}^2}$, R = sphere radius. Continuum value at $\varepsilon = 80$, $R = 25 \text{ \AA}$: $C \approx 6.56$.

C2 • The net charge enclosed in *pure* state s

$$Q_s = Q_{\text{env}} + \sum_{iq} q_{iq}^{(s)}, \quad Q_{\text{env}} = \sum_{\substack{i \in \text{solute} \\ i \notin \text{excl}, i \notin Q}} q_i$$

Q_{env} (protein + ions, fixed) plus the Q-region charge *in that end state* (no λ -mixing). The quantity whose Born self-energy we correct.

C3 • The per-state Born addition – monopole baseline

$$E_Q(s) \text{ += } E_{\text{Born}}(Q_s) = -C Q_s^2$$

This constant must enter the **per-state** energy that BAR differences. It removes the leading missing-exterior monopole term, but the data show that it is a baseline rather than a complete artifact model. Q's existing **charge_correction** is different: it changes the water-restraint target and therefore the sampled ensemble, while its state dependence is absent from $E_Q(s)$.

C4 • What the correction does to $\Delta\Delta G$

$$\Delta\Delta G_{\text{Born}} = -C (Q_2^2 - Q_1^2) = \underbrace{-2C Q_{\text{env}} \Delta q}_{\text{dominant}} - C (q_2^2 - q_1^2)$$

Neutral edges ($Q_1 = Q_2$) $\rightarrow 0$: untouched, safe to leave on. **Charge edges** $\rightarrow$ dominant term linear in the environment charge and the ligand charge change Δq . Parameter-free.

C5 • The defensible analytic baseline: two legs, two radii

$$\Delta\Delta G_{\text{corr}} = \underbrace{\Delta G_{\text{Born}}^{\text{protein}}(R_p)}_{\text{complex leg}} - \underbrace{\Delta G_{\text{Born}}^{\text{water}}(R_w)}_{\text{free-ligand leg}}, \quad R_w \approx 24.7 > R_p \approx 23.0 \text{ \AA}$$

Apply the per-state Born to *both* FEP legs and subtract. Use each topology's **qprep-computed effective solvent radius**, the radius Q actually restrains; it is not a fitted “effective radius”. This is the parameter-free leading correction, not a claim that all residuals are Born-like.

C6 • Validation: the leading artifact is removed, residuals remain

$$\text{raw artifact } 40\text{--}120 \frac{\text{kcal}}{\text{mol}} \xrightarrow{\text{two-leg per-leg Born}} \text{RMSE} \sim 1.5\text{--}17$$

Post-hoc results (kcal/mol): eg5 1.5, cmet_r35 2.7, tyk2 3.3, cmet 3.9, jnk1 4.0, jak1 8.2, ptp1b 17.0. The

Q_{env} dependence validates the leading Q^2/R form; the target-dependent remainder shows why it cannot yet be called the final solution.

C7 • The `phi_cog` diagnostic: local, not exterior

$$\phi_{\text{cog}} = \sum_{j \in \text{env}} \frac{k_e q_j}{\|\mathbf{r}_{\text{cog}} - \mathbf{r}_j\|^2} \quad \Delta G_{\text{monopole}} \approx \Delta q \langle \phi_{\text{cog}} \rangle_\lambda$$

This is the direct *in-sphere* environment potential at the ligand centroid under Q 's distance-dependent dielectric ($\varepsilon(r) = r \Rightarrow 1/r^2$). It is a useful force-free local diagnostic, but it is not a measurement of the missing exterior Born polarization. It does not yield a transferable residual correction.

C8 • The residual is scatter, not a law (honest limits)

residual $\sim$ per-edge scatter + per-target offset (no predictable Δq or ϕ term)

Two-leg per-leg Born is the parameter-free **baseline**. The leftover has **no** robust Δq -linear or $\Delta q\phi$ structure – tyk2 shows ~ 8 kcal of scatter at a *single* Δq , and ϕ_{pocket} is refuted (eg5 sign mismatch; cmet vs cmet_r35 move oppositely). The present data do not establish a universal residual sign or radius law. Possible contributions include higher boundary multipoles, water-model surface potential, and restraint-Hamiltonian inconsistency; separate them with controlled radius/off-centre tests rather than fitting an “effective radius” to experiment.

C9 • Make the charge-dependent restraint thermodynamically visible

$$H_m(\mathbf{x}) = \sum_s \lambda_s^m \left[V_s(\mathbf{x}) + W_{\text{pol}}^{(s)}(\mathbf{x}) - CQ_s^2 \right]$$

Current Q samples a λ -dependent water-polarization restraint but does not expose its pure-state energies to BAR. The consistent implementation evaluates one $W_{\text{pol}}^{(s)}$ per alchemical state, mixes their forces with λ_s^m , and stores each energy in $E_Q(s)$. Keep the Born term independently switchable, and freeze the learned restraint offset during production so every window samples a stationary Hamiltonian.